

A General Conditioning Diagnostic

David A. Belsley

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In this paper a conditioning diagnostic is suggested that can be applied quite generally to nonlinear estimators as well as to simultaneous-equations estimators. It is based on a generalization of the traditional collinearity diagnostics of Belsley, Kuh, and Welsch (1980) (BKW) which, with notable exceptions², are designed to be applied to strictly linear models (linear in both parameters and variates) estimated by ordinary least squares, that is, to the LS/linear case. Like those diagnostics, this procedure can become part of an active and passive guidance mechanism in a statistical computer environment.

Consider, then, the general model

$$g(\mathbf{Z}, \boldsymbol{\theta}, \mathbf{u}) = 0, \quad (1)$$

where g can be a vector-valued nonlinear stochastic function in the p parameters $\boldsymbol{\theta}$, $\mathbf{u}$ is an n -vector error term, and $\mathbf{Z}$ is an $n \times k$ matrix of k variates, which may be endogenous or exogenous. This formulation encompasses nonlinear and simultaneous-equations models. In this latter case, it is assumed that the system has been "stacked" into its single-equation equivalent with identifying restrictions.³ In the event that interest focuses directly on $\boldsymbol{\theta}$, we can rewrite (1) more simply as $g(\boldsymbol{\theta})$.

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² See Belsley (1988).

³ See Theil (1971, Chapter 10).

Let

$$\hat{\boldsymbol{\theta}} = \boldsymbol{\phi}(\mathbf{Z}) \quad (2)$$

be any consistent estimator of $\boldsymbol{\theta}$ with variance–covariance matrix $\mathbf{V}(\hat{\boldsymbol{\theta}})$, and let $\mathbf{S}(\hat{\boldsymbol{\theta}})$ be any consistent estimator of $\mathbf{V}(\hat{\boldsymbol{\theta}})$.

In the LS/linear case, (1) takes the familiar form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{V}(\hat{\boldsymbol{\theta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$, and $\mathbf{S}(\hat{\boldsymbol{\theta}}) = s^2(\mathbf{X}^T\mathbf{X})^{-1}$, where $s^2 = \mathbf{e}^T\mathbf{e}/(n-p)$. And in this case, as is shown in BKW, conditioning can be assessed by the scaled condition indexes $\tilde{\eta}_i(\mathbf{X})$ of $\mathbf{X}$, and variate involvement can be discerned through the variance–decomposition proportions of $\mathbf{X}$. The scaled condition numbers are derived as

$$\tilde{\eta}_k \equiv \frac{\mu_{\max}}{\mu_k} \quad k = 1, \dots, p, \quad (3)$$

where the μ_k are the singular values of $\mathbf{X}$ and are obtained from the singular–value decomposition of $\mathbf{X}$ into $\mathbf{UDV}^T$, where $\mathbf{U}^T\mathbf{U} = \mathbf{V}^T\mathbf{V} = \mathbf{I}_p$, and $\mathbf{D} \equiv \text{diag}(\mu_1, \dots, \mu_p)$, and where $\mathbf{X}$ has first been scaled so that the Euclidean length of each of its columns is unity. The variance–decomposition proportions π_{jk} are then derived by first calculating

$$\phi_{kj} \equiv \frac{v_{kj}^2}{\mu_j^2} \quad \text{and} \quad \phi_k \equiv \sum_{j=1}^p \phi_{kj} \quad k = 1, \dots, p, \quad (4)$$

and then, forming

$$\pi_{jk} \equiv \frac{\phi_{kj}}{\phi_k}, \quad k, j = 1, \dots, p. \quad (5)$$

Equivalently, these diagnostics can be determined from $\mathbf{X}^T\mathbf{X}$. Recalling that $\mu_i = \lambda_i^{-1/2}$, that is, the singular values of $\mathbf{X}$ are simply the positive square roots of the eigenvalues of $\mathbf{X}^T\mathbf{X}$, the scaled condition indexes are simply the ratios $(\lambda_{\max}/\lambda_i)^{-1/2}$,

and the variance–decomposition proportions are derived by applying (4) and (5) directly to the matrix $\mathbf{V}$ of the eigenvectors of $\mathbf{X}^T\mathbf{X}$.

Thus, in the LS/linear case, the conditioning diagnostics are obtainable from $\mathbf{S}^{-1}(\mathbf{b}) = s^{-2}\mathbf{X}^T\mathbf{X}$, after $\mathbf{X}$ has been column–equilibrated, so that the diagonal elements of $\mathbf{X}^T\mathbf{X}$ are all equal. The scale factor of s^{-2} is irrelevant since it cancels from all the ratios determining the condition indexes and the variance–decomposition proportions.

A Suggested Diagnostic

This, then, suggests using $\mathbf{S}^{-1}(\hat{\boldsymbol{\theta}})$ rather generally as a basis for a conditioning diagnostic. Specifically, relative to (1) and (2), consider the following:

STEP 1. Obtain $\hat{\boldsymbol{\theta}}$ and its estimated variance–covariance matrix $\mathbf{S}(\hat{\boldsymbol{\theta}})$.

STEP 2. Scale $\mathbf{S}^{-1}(\hat{\boldsymbol{\theta}})$ so that it has unit diagonal elements. This matrix, which we denote $\mathbf{W}$, is derived as $\mathbf{W} \equiv \mathbf{D}^T \mathbf{S}^{-1}(\hat{\boldsymbol{\theta}}) \mathbf{D}$, where $\mathbf{D} \equiv \text{diag}(d_1 \cdot \cdot \cdot d_p)$, and d_i is the inverse of the positive square root of the i –th diagonal element of $\mathbf{S}^{-1}(\hat{\boldsymbol{\theta}})$.

STEP 3. Obtain the eigenvalues λ_i and eigenvectors $\mathbf{V}$ of $\mathbf{W}$.

STEP 4. Form the condition indexes $\tilde{\eta}_i = (\lambda_{\max}/\lambda_i)^{-1/2}$, $i = 1, \dots, p$, and the variance–decomposition proportions as in (4) and (5), with $\mu_i = \lambda_i^{-1/2}$.

This procedure is clearly a proper generalization of the collinearity diagnostics of BKW, reducing to them in the LS/linear case. The results of STEP 4 can be brought together into a Π matrix identical in form to that described there. Care, however, must be taken in interpreting just what these figures mean, and we will be better able to see this after examining a special case.

Gauss-Newton with a Linearized Model

In the case that (1) takes the often-specified form

$$\mathbf{g}(\mathbf{Z}, \boldsymbol{\theta}) = \mathbf{u}, \quad (5)$$

$\boldsymbol{\theta}$ can be estimated through a minimum-distance (least-squares) approach which finds the $\hat{\boldsymbol{\theta}}$ that, given $\mathbf{Z}$, minimizes $\mathbf{u}^T \mathbf{u} = \mathbf{g}^T(\boldsymbol{\theta}) \mathbf{g}(\boldsymbol{\theta})$. One method for obtaining this solution is that of Gauss-Newton, using the first two terms of a Taylor expansion of (5) to form a linear approximation to the model about $\boldsymbol{\theta}^\circ$ as

$$\mathbf{z} = \mathbf{J}(\boldsymbol{\theta}^\circ) \boldsymbol{\theta} - \mathbf{u}, \quad (6)$$

where $\mathbf{z} \equiv \mathbf{J}(\boldsymbol{\theta}^\circ) \boldsymbol{\theta}^\circ - \mathbf{g}(\boldsymbol{\theta}^\circ)$ and $\mathbf{J}(\boldsymbol{\theta}^\circ) \equiv \left(\frac{\partial \mathbf{g}}{\partial \boldsymbol{\theta}^\circ} \right)$, the $n \times p$ Jacobian matrix of $\mathbf{g}$ with respect to $\boldsymbol{\theta}$ evaluated at $\boldsymbol{\theta}^\circ$.⁴ It is well known that iteratively re-estimating (6) with least squares as $\boldsymbol{\theta}_r = (\mathbf{J}^T(\boldsymbol{\theta}_{r-1}) \mathbf{J}(\boldsymbol{\theta}_{r-1}))^{-1} \mathbf{J}^T(\boldsymbol{\theta}_{r-1}) \mathbf{z}_r$ results in $\hat{\boldsymbol{\theta}}$.⁵ Further the estimated variance-covariance matrix at this solution is simply $s^2(\mathbf{J}^T(\hat{\boldsymbol{\theta}}) \mathbf{J}(\hat{\boldsymbol{\theta}}))^{-1}$, which, as long as the derivatives in $\mathbf{J}$ are predetermined, can be shown to be a consistent estimator of $\mathbf{V}(\hat{\boldsymbol{\theta}})$. Thus, asymptotically, any $\mathbf{S}(\hat{\boldsymbol{\theta}})$ used in the diagnostic suggested above is approximated by $s^2(\mathbf{J}^T(\hat{\boldsymbol{\theta}}) \mathbf{J}(\hat{\boldsymbol{\theta}}))^{-1}$, and the conditions of estimation are approximated linearly by

$$\mathbf{z} = \mathbf{J}(\hat{\boldsymbol{\theta}}) \boldsymbol{\theta} - \mathbf{u}. \quad (7)$$

Interpreting the Diagnostics

The preceding means that, in the context of (5), the diagnostics obtained by the procedure suggested above are approximations to the standard collinearity diagnostics applied to (7). That is, they tell us something about the conditioning of the pseudodata

⁴ See Bard (1974) or Judge *et al.* (1982, Chapter 24).

⁵ See Judge *et al.* (1982, Chapter 24).

that comprise the Jacobean matrix $J(\hat{\theta})$, specifically, the number and strengths of the near dependencies that exist in $J(\hat{\theta})$ and the involvement of the various columns of $J(\hat{\theta})$ in those near dependencies.

These pseudodata may or may not be structurally interpretable, that is, be in a form in which changes in the data can be assessed as being consequential or not relative to the underlying real-life situation they describe. It is beyond the current paper to discuss this concept in detail, but we note that it is shown in Belsley (1984), (1986) and in Belsley and Oldford (1986) that the data must possess structural interpretability if conditioning analyses are to be meaningful. Since the pseudodata are derivatives of the various nonlinear relations involving the Z variates which are themselves likely to be the structurally interpretable forms, it can be shown that it is unlikely that they will themselves be structural interpretable.

Thus, care must be taken in interpreting the diagnostic information obtained from $J(\hat{\theta})$. However, this information is by no means useless. For we have seen that estimation of (5) is akin to estimating (7) by least squares, and hence data weaknesses, however they arise, that can degrade this least-squares estimation can also affect all estimators and are worth knowing about.

Here, data weakness does not refer, for example, to collinearity or short-data⁶ among the basic Z variates, but rather to collinearity or short-data among the columns of $J(\hat{\theta})$, the result of the particular nonlinear transformations that Z undergoes in the formulation (5). Thus, the individual variance-decomposition proportions in the Π matrix cannot be associated with a specific Z variate, but they still can be associated with a specific estimated parameter variance. Even so, high variance-decomposition proportions linked up with high condition indexes are still symptomatic that the information in the Z data, after undergoing the transformations imposed by the given

⁶ See Belsley (1982) if further reference is needed for this term.

model, are unable to be used effectively for estimating particular of its parameters. Furthermore, if it is possible to associate particular $\mathbf{Z}$ s with particular columns of $\mathbf{J}(\hat{\boldsymbol{\theta}})$ that are involved in near dependencies, then it becomes possible to see more clearly why the existing $\mathbf{Z}$ data are inadequate and what corrective steps are most likely to prove useful.

The magnitudes of the scaled condition indexes that result from the suggested diagnostic can be more meaningfully interpreted as diagnostics for data conditioning if information can be obtained regarding structurally interpretable changes in the columns of $\mathbf{J}(\hat{\boldsymbol{\theta}})$. Let us suppose that the $\mathbf{Z}$ s are structurally interpretable, so that we can determine a set $\Omega_{\mathbf{Z}}$ of reasonable perturbations. Further, let us write $\mathbf{J}(\hat{\boldsymbol{\theta}})$ as $\mathbf{J}(\hat{\boldsymbol{\theta}}, \mathbf{Z})$ to highlight its dependence on the basic $\mathbf{Z}$ data. Then, perturbations $\delta\mathbf{Z} \in \Omega_{\mathbf{Z}}$ map into a set $\Omega_{\mathbf{J}}$ of induced reasonable perturbations in $\mathbf{J}(\hat{\boldsymbol{\theta}}, \mathbf{Z})$, namely, $\Omega_{\mathbf{J}} = \{\delta\mathbf{J} \equiv \mathbf{J}(\hat{\boldsymbol{\theta}}, \mathbf{Z} + \delta\mathbf{Z}) - \mathbf{J}(\hat{\boldsymbol{\theta}}, \mathbf{Z}) : \delta\mathbf{Z} \in \Omega_{\mathbf{Z}}\}$. It is the implied relative changes in the set $\Omega_{\mathbf{J}}$ that are best used in interpreting the magnitude of the scaled condition indexes.

Finally, it should be noted that the interpretative considerations derived above from the Gauss–Newton approximation technically hold only if the Jacobian matrix $\mathbf{J}(\hat{\boldsymbol{\theta}})$ is predetermined, that is, only if the derivatives in $\mathbf{J}(\hat{\boldsymbol{\theta}})$ are not functions of currently endogenous variates. In the event that several of the $\mathbf{Z}$ s are currently endogenous and are associated with parameters to be estimated, this will not be true, and this interpretation is no longer strictly valid. It remains to be seen, however, if this very valid statistical consideration has any serious affect on the diagnostic value of the suggested procedure.

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